

fraction = 0.18

A1 [2]

- 7 (a) (i)** quantum/packet/discrete amount of energy
electromagnetic mentioned
- (ii)** max. k.e. corresponds to electron emitted from surface
energy is required to bring electron to surface
- (b)** at higher frequency, fewer photons (per second) for same intensity
so rate of emission decreases
(allow argument based on photoelectric efficiency)

M1
A1 [2]

B1
B1 [2]

M1
A1 [2]

- 8 (a) (i)** either: number = $6.02 \times 10^{23} \times (2.65 \times 10^{-6}) / (22.4)$